
North Coast Regional Water Quality Control Board

May 16, 2016

Silver Oak Cellars Alexander Valley
Silver Oak Cellars
Attn: Tony Le Blanc, Chief Operating Officer
P.O. Box 558
Geyserville, CA 95448

Dear Mr. Le Blanc:

Subject: Silver Oak Cellars Alexander Valley Domestic Wastewater Treatment System,
Geyserville, Sonoma County – Notice of Applicability of Water Quality Order
WQ 2014-0153-DWQ

File: Silver Oak Cellars, Highway 128 Alexander Valley (Domestic Waste), 7300
Highway 128, Geyserville, CA; WDID No. 1B15087RSON; CIWQS Place No.
816488

This letter serves as Notice of Applicability (NOA) from the North Coast Regional Water Quality Control Board (Regional Water Board) that the General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems, Order WQ 2014-0153-DWQ (General Order) is applicable to the subject wastewater system and that the system is henceforth regulated under the General Order as described below. A Monitoring and Reporting Program (MRP) for the wastewater system is included as an enclosure of this NOA.

Silver Oak Cellars (Discharger) submitted a report of waste discharge (ROWD) dated June 25, 2015, for the proposed Silver Oak Cellars Domestic Wastewater Treatment System located at 7300 Highway 128, Geyserville in Sonoma County. Supplemental information regarding groundwater mounding and nitrate loading was provided on July 1, 2015. Based on the information presented in the ROWD, the proposed wastewater treatment system and discharge are consistent with the requirements of the General Order.

The General Order contains requirements that apply to all wastewater systems and also requirements that apply only to specific types of treatment and/or disposal systems. Applicable requirements and effluent limitations are discussed in Section 3 of the NOA and

summarized in Table 1, an enclosure of this NOA. The General Order contains additional requirements that are not summarized in Table 1. The Discharger is responsible for all applicable requirements in the General Order, this NOA, and the MRP.

A copy of the General Order is enclosed with this NOA. The General Order is also available on the Internet at:

http://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wqo2014_0153_dwq.pdf

1. Facility and Discharge Description

Facility

The Silver Oak Cellars - Alexander Valley winery is owned and operated by Silver Oak Cellars and is located in Geyserville, Sonoma County. The winery is not located in an area served by a regional wastewater collection system; therefore, domestic and process wastewater is collected and treated on-site. The site location is shown on Attachment A, which is attached hereto and is made part of this NOA by reference.

The Silver Oak Cellars - Alexander Valley winery facility (Facility) is located on a 115-acre parcel (APN No. 131-200-020) and consists of: a production and administration building; a storage building; a wine tasting building; a marketing accommodations (MA) three bedroom residence; associated utilities and infrastructure; and vineyards. The winery is currently permitted with a maximum annual wine production capacity of 120,000 cases, as authorized by Sonoma County Use Permit PLP14-0004.

Two separate wastewater treatment systems will serve the Facility. Domestic wastewater originating from facility restrooms, laboratories, the tasting room, and the MA residence will be treated and disposed of via the domestic wastewater treatment system. The domestic wastewater system will consist of a gravity collection system, two septic tanks, one kitchen grease interceptor, an Orenco Advantex AX-MAX125 aerobic biological filter unit (with recirculation and dosing tanks built-in), an ultra-violet (UV) light disinfection system, and a subsurface drip disposal field. Following pretreatment and disinfection, the treated domestic wastewater will be disposed of through a subsurface drip disposal field.

The Facilities winery process wastewater will be treated via a Membrane Bioreactor (MBR) package wastewater treatment system for reuse as vineyard irrigation water or non-potable process reuse water. The process wastewater discharge will need to be authorized under General Waste Discharge Requirements (WDR) for wine, beverage and food processor waste, under Conditional Waiver of WDR for wine, beverage and food processor waste discharges, or under an individual order. This notice of applicability of the General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems, Order WQ 2014-0153-DWQ (General Order) is applicable to the domestic wastewater system and not the winery process wastewater system.

Discharge Description

The domestic wastewater treatment system is designed for a peak daily flow of 4,025 gallons per day (gpd). Peak daily flow conditions are predicted to take place during crush when the maximum number of employees are present, the tasting room attendance is at a

peak, and a 200 guest special event is taking place. The average daily wastewater flow is projected to be 1,575 gpd during non-crush periods. As described in the ROWD, the projected untreated and treated domestic wastewater quality from the Facility is consistent with the characterization of wastewater in the General Order.

The overall plan layout of the wastewater collection system is included in the *Sanitary Sewage Management System Improvements* design plans and the *Design Calculations* report, for Silver Oak Cellars, prepared by Summit Engineering Inc., and dated July 15, 2014, and May 8, 2015, respectively. These plans and report are made part of this NOA by reference.

Primary Treatment and Collection System

Primary domestic waste treatment and collection will be provided using a standard gravity collection system, kitchen grease interceptor and septic tanks, as described below.

A 1,500-gallon precast concrete septic tank with effluent filter will serve the MA residence. The septic tank is designed to provide a 3-day detention time for solids removal prior to effluent entering the larger winery septic tank. A 13,000-gallon fiberglass septic tank with effluent filter will serve the entire Facility. The 13,000-gallon septic tank is designed to provide a 3-day detention time for solids removal and to act as a sump prior to transferring effluent to the Advantex pretreatment system. The 13,000 gallon septic tank will be equipped with duplex pumps and float switches that will be controlled and monitored by an adjacent pump control panel with visual and audible alarm.

The tasting room and hospitality building will be served by a 1,500-gallon kitchen grease interceptor, designed for a maximum flow of 500 gpd and a 3-day retention time. The precast concrete grease interceptor is intended to remove oil and grease from the kitchen wastewater prior to entering the septic tank.

Secondary Wastewater Treatment System

A wastewater treatment system, consisting of an Advantex recirculating aerobic biological filter system, will be installed to provide secondary level treatment for the domestic wastewater flow. The treatment system consists of one filter unit, a recirculation tank, dosing tank, piping, pumps, and controls. The filter unit, also referred to as a treatment pod, is a manufacturer-supplied modular unit consisting of a fiberglass tank containing textile filtration media, a pressurized influent distribution pipe network, and an underdrain to collect treated wastewater. The secondary level treatment is designed to reduce the organic strength and solids content of the domestic wastewater prior to disposal to the subsurface drip disposal field. The treatment system will be equipped with a pump control panel with remote telemetry that will control and monitor the pumps associated with the treatment system and the subsurface drip disposal field. A central remote alarm panel will be located near the winery processing area.

Wastewater Discharge System

Following secondary level treatment, effluent will be disinfected and disposed of at a subsurface drip disposal field. A flow meter will measure the volume of effluent being sent to the disposal field. A UV disinfection basin outfitted with a UV unit rated for 83 gallons per minute (gpm) will provide disinfection prior to subsurface disposal. The UV system

manufacturers cited water quality recommendation of a maximum concentration of 10 mg/L suspended solids prior to UV light exposure.

A subsurface drip disposal field will be located in a vineyard on the north end of the parcel near Sausal Creek. The disposal field is designed to dispose of 4,025 gpd of effluent with an application rate of 1.2 gallons/square foot/day. The field will consist of eighteen 100-foot sections of Geoflow drip lines, spaced two feet apart, at a depth of 12-18 inches below ground surface. A 200% reserve disposal field area has been identified in the same area, directly adjacent to the subsurface disposal field.

The disposal field, reserve area, and treatment system components meet the set back requirements in the General Order.

General WDR Order Coverage

Commencing on the date of this NOA, waste discharges from the Facility's domestic wastewater treatment system are regulated under the General Order (Order WQ 2014-0153-DWQ).

2. California Environmental Quality Act

On October 9, 2014, the Sonoma County Board of Zoning Adjustments approved a Negative Declaration under the California Environmental Quality Act (CEQA) for the construction/expansion of Silver Oak Cellars – Alexander Valley winery and the subject domestic wastewater system (State Clearing House #2014082047). No water quality related mitigation measures were included as a condition of project approval.

3. Site-Specific Requirements and Effluent Limitations

A. General

The General Order contains prohibitions and specifications that apply to all wastewater systems covered under the General Order, and specifications that apply only to specific types of treatment and/or disposal systems. Table 1 of this NOA provides a summary of applicable discharge requirements for the subject wastewater system and associated effluent limitations. Discharge of wastes other than those described in the ROWD is prohibited.

Wastewater system managers and operators must be familiar with the applicable requirements contained in the General Order, this NOA, and MRP R1-2016-0026.

B. Reports to be Submitted to the Regional Water Board

1. The following technical reports, which form the main components of a comprehensive Operation, Maintenance, and Monitoring program, shall be submitted as described below:
 - a) By August 16, 2016, the Discharger shall submit a Spill Prevention and Emergency Response Plan (Response Plan) consistent with the requirements of General Order Provision E.1.a. The Response Plan shall, at a minimum, address the following:

- 1) Operation and Control of Wastewater Treatment
- 2) Sludge Handling
- 3) Collection System Maintenance
- 4) Emergency Response
- 5) Notification Procedures

b) By August 16, 2016, the Discharger shall submit a Sampling and Analysis Plan consistent with the requirements of General Order Provision E.1.b.

2. The above reports shall be acceptable to the Regional Water Board's Executive Officer.

C. Septic Tanks

The domestic wastewater system serving the Facility includes septic tanks that provide primary treatment of the wastewater. Septic tanks require proper operation, maintenance, and monitoring in order to provide adequate and reliable wastewater service. This includes periodic inspection and removal, as needed, of accumulated solids. Requirements for septic tanks are included in General Order section B.2, Septic Systems, and in the enclosed MRP, at section titled "Septic Tank Monitoring and Maintenance." Operational maintenance and monitoring work on the septic systems shall be conducted by qualified service providers with appropriate oversight by the Discharger. The Discharger is responsible for ensuring proper operation, maintenance, and monitoring of all septic tanks.

D. Subsurface Disposal System Requirements

The domestic wastewater system includes dispersal of treated effluent to subsurface soil via a 3,600 square foot subsurface drip disposal system. The dispersal system will include eighteen, 100 lineal-foot drip lines, spaced two feet apart, at 12-18 inches below ground surface. The Discharger is required to include a constructed primary dispersal system with capacity for 100 percent of design flows, and a 100-percent capacity designated reserve area. The reserve area is for future use, in case the existing disposal system is found to be inadequate. The reserve area is required to remain available until needed. The Discharger has designated a 200-percent capacity (7,200 square foot) reserve disposal area adjacent to the primary disposal area along its northeast boundary.

By August 16, 2016 the Discharger shall submit "As-Built" Plans for the constructed subsurface disposal system and associated monitoring wells.

E. Effluent Limitations

Effluent limitations for biochemical oxygen demand (BOD) and total suspended solids (TSS) are presented in Table 1. **The effluent limitation for BOD is 30 mg/L. The effluent limitation for TSS is 30 mg/L.** The numeric value of the effluent limitation for BOD and TSS are consistent with performance-based effluent limitations specified in Table 4 of the General Order for an aerobic biological filter system. Effluent limitations for BOD and TSS are expressed as monthly average limitations and are calculated by taking the arithmetic mean of all measurements collected in a calendar month.

Regional Water Board staff evaluated the need for a nitrogen effluent limitation using the methodology contained in the General Order and determined a nitrogen effluent limitation is not required because the treatment system flow rate is less than 20,000 gpd.

F. APMP Geographic Area Eligibility

The geographic area of an APMP is established (in order of hierarchy) by an approved TMDL, an approved local agency defined APMP, or 600 feet from a waterbody listed in Attachment 2 of the OWTS Policy. In general, wastewater systems located within the geographic area of an APMP are not eligible for coverage under this General Order unless the Regional Water Board's Executive Officer determines the discharge is acceptable based on site-specific conditions, the level of wastewater treatment, and/or total maximum daily load (TMDL) implementation requirements.

The facility is not within the area of an Advanced Protection Management Program (APMP). In the event that the domestic wastewater system is found to be located within the geographic area of an Advanced Protection Management Program (APMP) for Pathogen Impaired Areas, the system will be evaluated for continued eligible coverage under the General Order.

4. Monitoring and Reporting Program

The Discharger shall comply with enclosed MRP R1-2016-0026, for the Silver Oak Cellars – Alexander Valley Domestic Wastewater Treatment System, which is made part of this NOA by reference.

The MRP includes requirements for the monitoring and maintenance of the septic tanks, monthly monitoring of the disposal field, and quarterly effluent monitoring.

Quarterly self-monitoring reports and an annual summary report are to be submitted to the Regional Water Board in accordance with the Table 5 of the MRP. The MRP should be consulted for all specific requirements.

5. Enforcement

Silver Oak Cellars shall manage and discharge the waste subject to the terms and conditions of the General Order and this NOA and shall maintain exclusive control over the discharges. As such, Silver Oak Cellars is responsible for compliance with the General Order and this NOA.

Failure to comply with the requirements in the General Order, this NOA, or the MRP could result in an enforcement action as authorized by provisions of the California Water Code.

California Water Code section 13267 states, in part:

“In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to

discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports.”

California Water Code section 13268 states, in part:

“(a) Any person failing or refusing to furnish technical or monitoring program reports as required by subdivision (b) of section 13267, or failing or refusing to furnish a statement of compliance as required by subdivision (b) of section 13399.2, or falsifying any information provided therein, is guilty of a misdemeanor and may be liable civilly in accordance with subdivision (b).

(b)(1) Civil liability may be administratively imposed by a regional board in accordance with article 2.5 (commencing with section 13323) of chapter 5 for a violation of subdivision (a) in an amount which shall not exceed one thousand dollars (\$1,000) for each day in which the violation occurs.”

The reports required in item 3.B are necessary to ensure that the Discharger complies with the NOA and General Order. Pursuant to California Water Code section 13267, the Discharger shall submit the reports described herein.

Please review this NOA carefully to ensure that it completely and accurately reflects the discharge. If the discharge violates the terms or conditions listed, the Regional Water Board may take enforcement action, including assessment of administrative civil liability. Prior to allowing changes to the wastewater strength or generation rate, or if the method of waste discharge changes from that described in the ROWD, you must contact the Regional Water Board to determine if submittal of an ROWD is required.

6. Annual Fees

The waste discharge authorized by this NOA is subject to payment of annual fees under the State Water Board’s annual fee schedule for regulated discharges. Annual fees are based in part on ratings for threat to water quality and complexity. The subject Facility is currently rated 3-C. The annual fee for this rating in accordance with the current (FY 2015-2016) Fee Schedule is \$2,088; however, based on the flow rate, the discharge qualifies for the 50-percent fee discount. The annual fee for the discharge is currently \$1,044.

The Fee Schedule is subject to change, and the fee for this Facility may change accordingly. Annual fees are due and payable as invoiced annually to the Discharger by the State Water Board.

7. Regional Water Board Contact Information

Regulation of the subject domestic wastewater treatment system under the General Order is managed by the Regional Water Board’s Groundwater Permitting Unit. The case is

currently assigned to Regional Water Board staff environmental scientist Rachel Prat. Questions or concerns about this NOA, the attached MRP, or related matters should be directed to Rachel Prat at (707) 576-2542 or Rachel.Prat@waterboards.ca.gov.

Sincerely,



Matthias St. John
2016.05.16
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Water Boards
Matthias St. John
Executive Officer

160516_RSP_ef_SilverOakAlexanderValley_WWTF_NOA

Attachments:

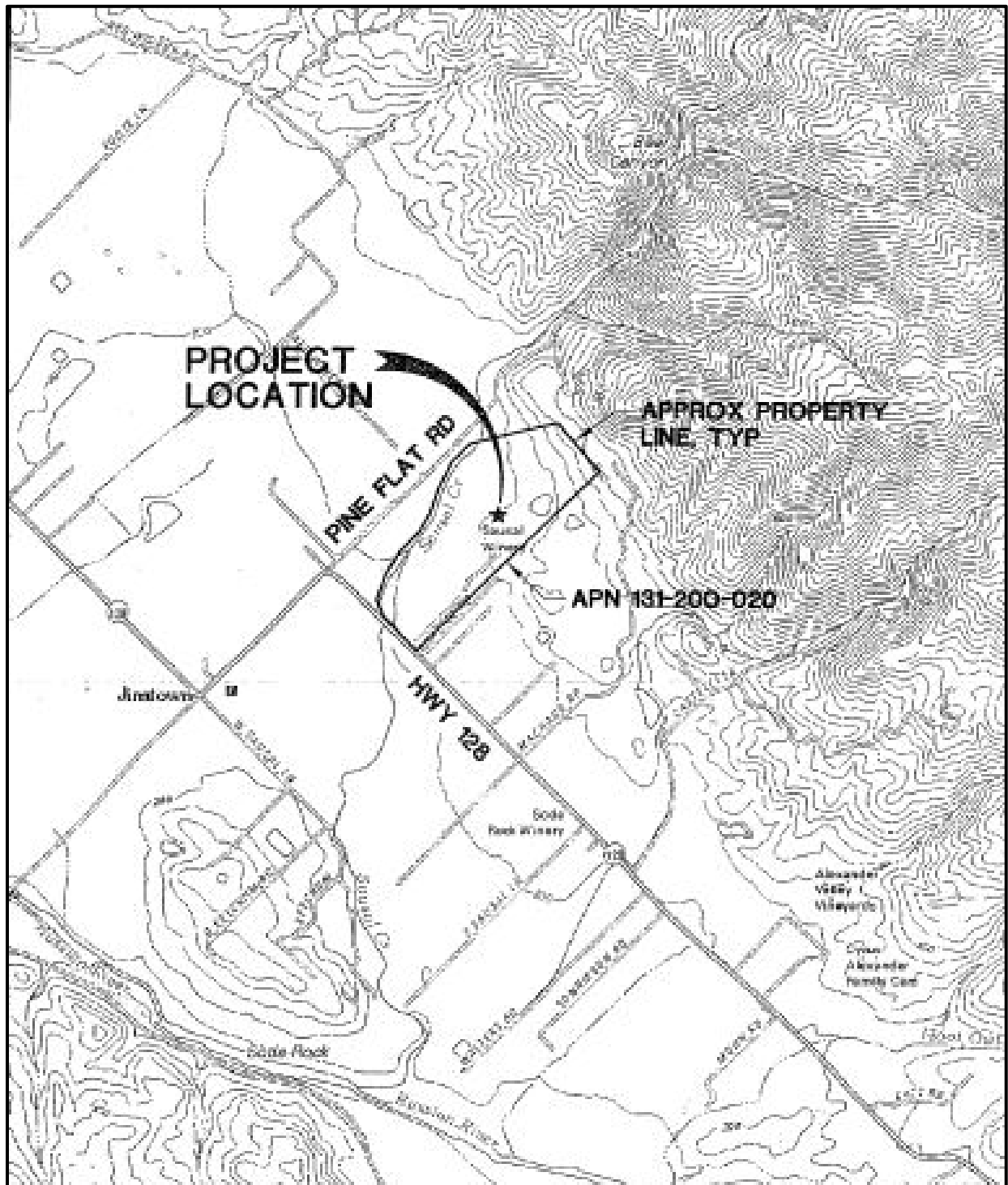
Attachment A, Site Location Map
Attachment B, Project Area Map
Attachment C, Wastewater Treatment System Schematic

Enclosures:

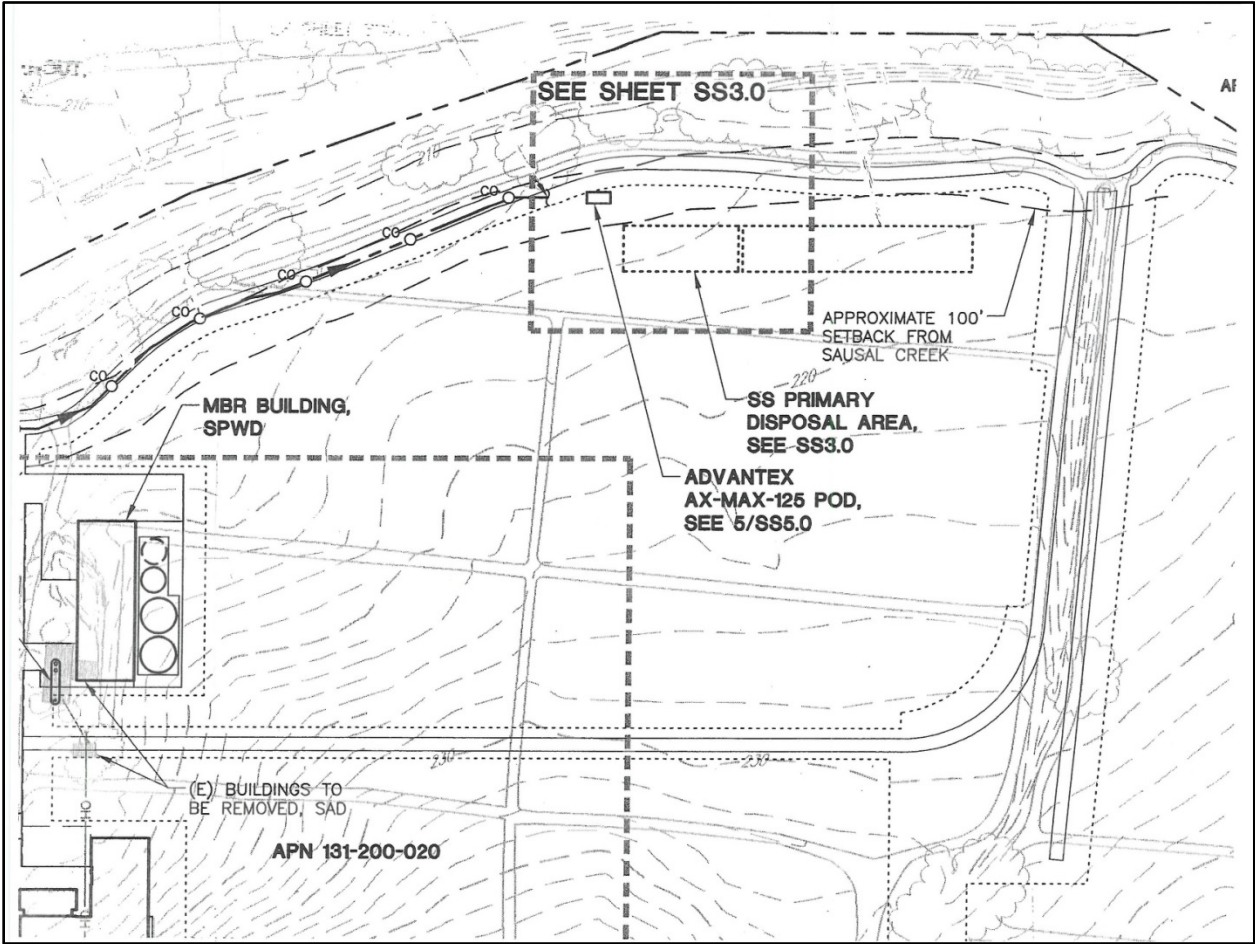
Table 1, Summary of Applicable Discharge Requirements
Water Quality Order WQ 2014-0153-DWQ
Monitoring and Reporting Program R1-2016-0026

cc Timothy O'Brien, State Water Resources Control Board, Sacramento,
Timothy.OBrien@waterboards.ca.gov
James Johnson, Sonoma County PRMD, Santa Rosa,
James.Johnson@sonoma-county.org
Claudia Llerandi, Summit Engineering, Inc., Santa Rosa
claudia@summit-sr.com

Attachment A – Site Location Map



Attachment B - Project Area Map



Attachment C – Wastewater Treatment System Schematic

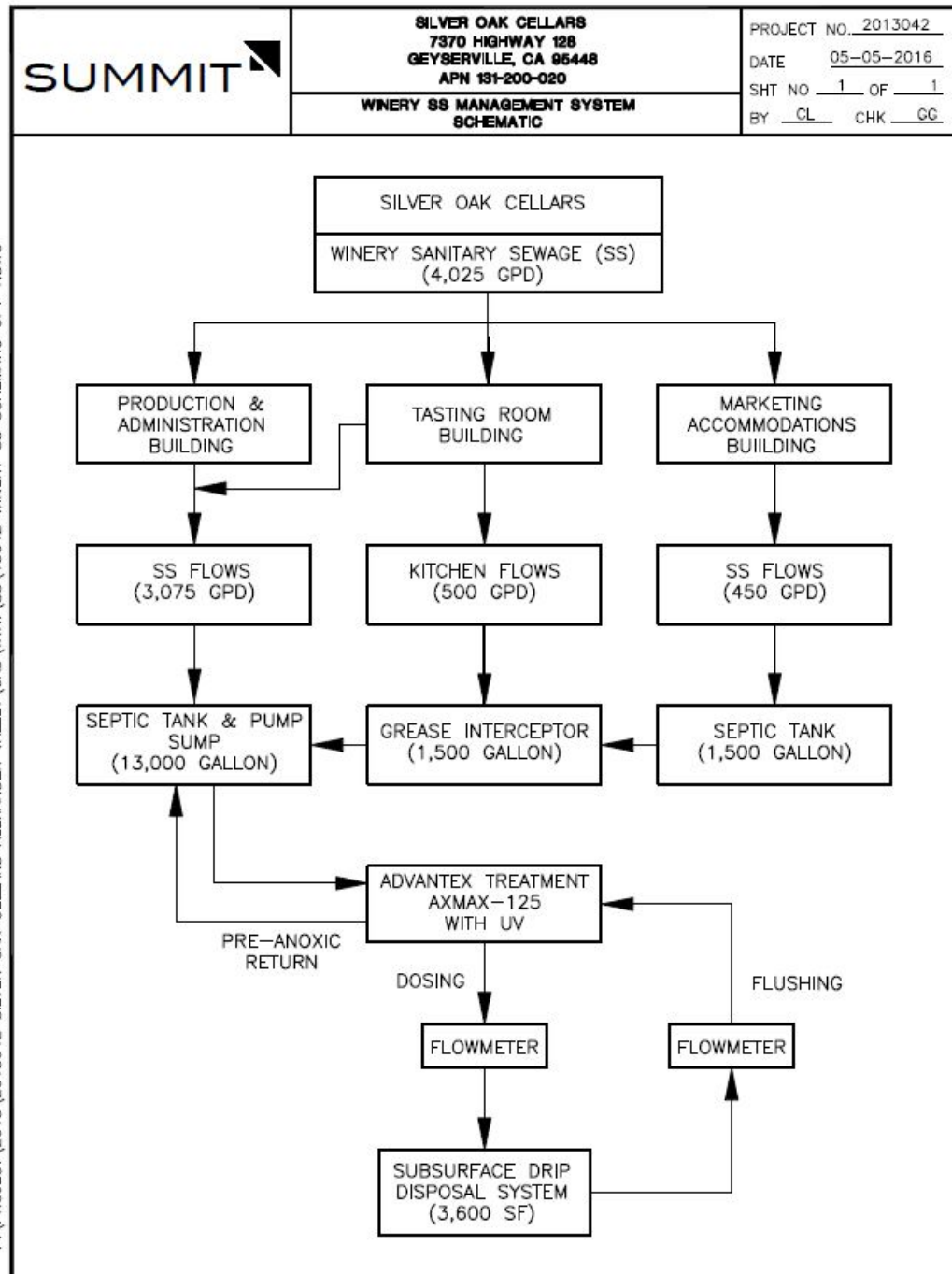


Table1, Summary of Facility Specific General Order Requirements (NOA Enclosure)

General Order Section	Page	Numeric Limitation	Requirement (refer to General Order and NOA for details)
A. PROHIBITIONS			
A.1-A.11	15	--	Prohibitions
B. REQUIREMENTS BY WASTEWATER SYSTEM TYPE			
B.1.a-B.1.l	16	--	All Wastewater Systems
B.1.a	16	4,025 gpd	Wastewater Flow Limit.
B.3.a-B.3.g	21	--	Aerobic Treatment Systems
B.6.a-B.6.h	24	--	Subsurface Disposal System
B.8.a-B.8.f	26	--	Sludge/Solids/Biosolids Disposal
C. GROUNDWATER AND SURFACE WATER LIMITATIONS			
C.1	27	--	Groundwater and Surface Water Limitations
D. EFFLUENT LIMITATIONS			
D.1.a	27	30 mg/L	Five Day Biochemical Oxygen Demand (BOD ₅), monthly average.
D.1.a	27	30 mg/l	Total Suspended Solids (TSS), monthly average.
E. PROVISIONS			
Technical Report Requirements			
E.1.a	28	Have on-site.	Spill Prevention and Emergency Response Plan
E.1.b	29	Have on-site.	Sampling and Analysis Plan
E.1.c	30	Not Applicable	Sludge Management Plan
E.2.h	32	Have on-site	Copy of NOA, and General Order